

Dog Breed Identification Using Transfer Learning

Project Overview

Dog Breed Identification is a deep learning-based image classification project that identifies the breed of a dog from an uploaded image using Transfer Learning.

The model leverages pre-trained convolutional neural networks such as ResNet50, MobileNetV2, or VGG16 trained on ImageNet to improve accuracy and reduce training time.

This project can be deployed as a web application where users upload dog images and receive predicted breed results.

Functional Requirements

- User can upload a dog image.
- System validates image format (JPG, PNG).
- Image preprocessing (resize, normalization).
- Model predicts dog breed.
- Display predicted breed with confidence score.
- Optional: Show top 3 predictions.
- Admin can retrain model with new data (optional feature).

Non-Functional Requirements

- High prediction accuracy (>85%)
- Fast response time (<3 seconds)
- Scalable deployment
- Mobile-friendly UI
- Secure image upload handling

Customer Journey

1. User visits the website.
2. User uploads a dog image.
3. System processes and validates image.
4. Model predicts breed.
5. System displays predicted breed with confidence score.
6. User can upload another image.

Data Flow

User → Upload Image → Frontend → Backend Server →
Image Preprocessing → Pre-trained CNN Model → Prediction → Display Result

Solution Architecture

Model Layer: Pre-trained CNN (ResNet50 / MobileNetV2), Fine-tuned layers, Softmax output.
Backend Layer: Flask or FastAPI for REST API.
Frontend Layer: HTML/CSS/JS or Streamlit.
Deployment Layer: AWS / Render / Docker.

Technology Stack

Programming Language: Python
Deep Learning Framework: TensorFlow / Keras / PyTorch
Backend: Flask / FastAPI
Frontend: HTML / CSS / JavaScript / Streamlit
Deployment: AWS / Render / Docker
Version Control: Git / GitHub

Expected Outcomes

- Accurate classification of 100+ dog breeds.
- Real-time prediction.
- User-friendly web application.
- Scalable AI solution.

